POWERACT CONSULTING

NCP Solver

DynamicMS Suite

SOFTWARE REQUIREMENTS SPECIFICATION

Functional & Non-Functional Requirements, and Interfaces

Version 1.0

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Confidential

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) captures the complete set of functional and non-functional requirements, and the external interfaces, of NCP Solver — DynamicMS Suite. It is derived from, and traceable to, the NCP Solver project scope, definition & design (PDD), information model, and knowledge-base source specifications, and from the delivered implementation itself. It is intended for product owners, developers, quality assurance, and auditors.

1.2 Scope

NCP Solver is a full-stack, multi-tenant, multi-language web application that manages non-conformities and problems (recorded as "NCP Sheets") through a structured seven-stage resolution process (E1-E7), augmented by rule-based AI agents, and surrounded by governance, risk, and compliance (GRC) modules: Business Rules, Controls (COSO framework), and Risks & Opportunities, plus an independently versioned AI Use Cases Library. The application supports both independent-company and group-of-companies tenancy models, role-based access control across nine standard roles, and English/French/Arabic localization including full right-to-left layout.

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
NCP Sheet	The central record of a non-conformity or problem, spanning stages E1-E7 (referred to as "NCP Fiche" in the original source specification).
E1-E7	The seven stages of the NCP resolution process: Detection, Understanding, Immediate Actions, Root Cause Analysis, Corrective Action Plan, Execution & Evaluation, Capitalization.
RR	Responsable Realisation - the user responsible for executing an action.
RE	Responsable Evaluation - the user responsible for verifying an action's effectiveness; must differ from the RR.
REX	Retour d'EXperience - the structured lessons-learned output of stage E7.
RCA	Root Cause Analysis - the discipline of finding the true underlying cause of a problem (5-Why, Ishikawa).
5W2H / QQOQCCP	Structured problem-description framework: What/Where/When/Who/Why/How/How-Much (French: Qui/Quoi/Ou/Quand/Comment/Combien/Pourquoi).
6M	Ishikawa cause categories: Man, Machine, Method, Material, Measurement, Milieu.
RAG	Retrieval-Augmented Generation - semantic search over the Capitalization Library grounding AI suggestions in the tenant's own historical data.
RBAC	Role-Based Access Control - access rights determined by assigned role(s).

Term	Definition
COSO	Committee of Sponsoring Organizations of the Treadway Commission - the internal control framework used to classify Controls.
OBS	Organizational Breakdown Structure - the site/department/service/team tree.
Multi-Tenant	Architecture in which multiple Organizations share one deployment with fully isolated data.
Composable Module	A module (Action Register, Alert Engine, RBAC, Audit Trail) built for reuse by other applications via a shared API contract.

1.4 References

- NCP Solver - Fast-Track Project Scope (source specification).
- NCP Solver - AI Project Definition & Design (PDD) v1.0.
- NCP Solver - Information Model (core classes / entity specification).
- NCP Solver - Knowledge Base for RAG v1.0 (KB-000 through KB-021).
- NCP-Solver: Global CIO Forum Use Case presentation, Fast-Track edition.
- POWERACT Consulting - Brand & Graphical Chart Guide v1.0.

1.5 Document Overview

Section 2 describes the product at a high level. Section 3 enumerates functional requirements by module, each individually identified (FR-xxx) for traceability. Section 4 defines external interfaces (user, software, and communications). Section 5 defines non-functional requirements (NFR-xxx). Section 6 summarizes the data model. Appendices provide the permission matrix, KPI formulas, alert catalog, and a full glossary.

2. Overall Description

2.1 Product Perspective

NCP Solver is a new, self-contained system composed of a React single-page-application frontend and a Node.js/Express REST API backend, persisting to a relational database. It is designed around six composable modules (Action Register, Capitalization Library/RAG, Alert & Notification Engine, User Management & RBAC, KPI Dashboard & Reporting, Audit Trail Service) intended for reuse by sibling enterprise applications (Risk Management, Audit Management, Project Management) via the same API contract.

2.2 Product Functions (Summary)

- Register, understand, contain, analyze, correct, execute/evaluate, and capitalize non-conformities via the E1-E7 NCP Sheet workflow.
- Manage an Action Register shared by immediate and corrective actions, with RR/RE-segregated effectiveness evaluation and evidence.
- Provide eight stage-specific AI agents that suggest — never autonomously decide — classifications, containment actions, root causes, corrective actions, and REX narratives.
- Maintain a semantically searchable Capitalization Library (RAG) of resolved problems, tenant-isolated.
- Maintain an independently versioned AI Use Cases Library with full labeled-section content history and revert.
- Maintain Business Rules, COSO-classified Controls, and a Risk/Opportunity register with a likelihood x impact heat map.
- Provide full role-based access control across 9 standard roles and a live permission matrix editor.
- Provide Group/Organization/Project hierarchy and an Organizational Breakdown Structure (OBS).
- Compute 10 KPIs and 4 standard reports, and raise 10 categories of alerts.
- Operate multi-tenant with complete data isolation, and multi-language with full Arabic RTL support.

2.3 User Classes and Characteristics

Role	Who They Are	Key Permissions
Administrator	IT/system administrator	Full CRUD on all records; user & RBAC management; tenant configuration; license & governance settings.
Quality Manager	Owns the quality management system	Validates and closes Sheets; configures KPI thresholds; owns Controls and Business Rules; compliance reporting.

Role	Who They Are	Key Permissions
CI Pilot	Coordinates the NCP process for their scope	Creates/edits/validates Sheets; approves action plans; manages REX; owns Business Rules and Risks/Opportunities in scope.
NCP Team Member	Cross-functional problem-solving team	Views/edits assigned Sheets; completes E2/E4/E5 stages; adds root causes and actions.
Action Owner (RR)	Executes an assigned action	Updates own action status; enters completion date; uploads execution evidence.
Evaluator (RE)	Verifies action effectiveness	Views pending evaluations; records effectiveness verdict; uploads evaluation evidence. Must differ from the action's RR.
Department Head	Manages an affected department	Views all Sheets in own department; department dashboard/reports; receives department-level alerts and risk visibility.
Reporter	Any employee authorized to report a problem	Creates new Sheets (E1 only); views own submitted Sheets; no edit rights after submission.
Read-Only / Auditor	Internal or external audit/compliance	Read-only access to Sheets, reports, Business Rules, Controls, Risks/Opportunities, and the full audit trail.

2.4 Operating Environment

Server: any Node.js 20+ runtime (Node 22 recommended), no external database server or third-party API key required (SQLite by default; the relational and RAG layers are each swappable for a distributed equivalent). Client: any evergreen desktop or tablet web browser. Deployment: SaaS (shared multi-tenant infrastructure) or OnPrem (customer-hosted), selectable per Organization.

2.5 Design and Implementation Constraints

- All AI agent behavior must be explainable and logged (agent name, input, output, confidence score); no black-box autonomous decisions.
- Every data access must be scoped by tenant (organization_id); no query may return another tenant's data.
- The UI must conform to the POWERACT Consulting visual identity across every module and every supported language.
- The RR and RE on any single action evaluation must always be different users.

2.6 Assumptions and Dependencies

- Users have modern web browsers with JavaScript enabled.
- Organizations are responsible for their own user provisioning within the tenant they administer.
- The rule-based AI agent implementations are a functional stand-in for a future LLM/vector-database-backed implementation, behind a stable interface.

3. System Features (Functional Requirements)

Each requirement is uniquely identified and stated in "the system shall" form for verifiability. Requirements are grouped by module in implementation order.

3.1 Dashboard

A role-aware landing page summarizing the authenticated user's open work, recent NCP Sheets, and unread alerts.

ID	Requirement
FR-DASH-01	The system shall display, on login, counts of NCP Sheets by status (Open, In Progress, Closed, Cancelled) scoped to the user's organization.
FR-DASH-02	The system shall display the count of the authenticated user's own open assigned actions and the count of organization-wide overdue actions.
FR-DASH-03	The system shall list the most recently created NCP Sheets, each showing sheet number, title, stage, criticality, and status.
FR-DASH-04	The system shall list the authenticated user's unread alerts with type, message, and timestamp.
FR-DASH-05	The system shall render a bar chart of non-conformity counts grouped by department (OBS node).
FR-DASH-06	The system shall provide a one-click action to create a new NCP Sheet, visible only to users holding the fiche.create permission.

3.2 Hierarchy Management (Group / Organization / Project)

Supports an optional Group (holding company) above the Organization (tenant) level, and optional Projects below it, per the Group-of-companies and independent-company deployment scenarios.

ID	Requirement
FR-HIER-01	The system shall allow an Organization to optionally belong to a Group; an Organization not assigned to a Group shall operate as fully independent.
FR-HIER-02	When an Organization belongs to a Group, the system shall display all sibling Organizations in that Group to authorized users.
FR-HIER-03	The system shall allow an authorized user to create an additional sibling Organization within the same Group.
FR-HIER-04	The system shall allow full CRUD on Projects (optional sub-entities of an Organization), including name, description, status, and start/end dates.
FR-HIER-05	The system shall allow an NCP Sheet to be optionally associated with a Project.
FR-HIER-06	The system shall allow an authorized user to edit core Organization attributes (name, sector, country).

3.3 Organizational Breakdown Structure (OBS)

A hierarchical site/department/service/team tree used to classify where a non-conformity occurred.

ID	Requirement
FR-OBS-01	The system shall support full CRUD on OBS nodes, each typed as Site, Department, Service, or Team.
FR-OBS-02	The system shall allow an OBS node to have a parent OBS node, forming an arbitrary-depth tree per Organization.
FR-OBS-03	The system shall render the OBS as an indented tree reflecting parent/child relationships.
FR-OBS-04	The system shall allow an NCP Sheet to be associated with exactly one OBS node (its department/origin).
FR-OBS-05	The system shall prevent deleting an OBS node from breaking referential integrity by cascading or reassigning dependent records per the data model.

3.4 Identity & Role-Based Access Control (RBAC)

User directory, the 9 standard roles from the NCP Solver Knowledge Base (KB-011), and a full permission matrix editor.

ID	Requirement
FR-ID-01	The system shall support full CRUD on users, including first/last name, email, username, language preference, and active status.
FR-ID-02	The system shall hash all user passwords (bcrypt) and never expose password hashes via any API response.
FR-ID-03	The system shall support assigning one or more of the 9 standard roles to a user: Administrator, Quality Manager, CI Pilot, NCP Team Member, Action Owner (RR), Evaluator (RE), Department Head, Reporter, Read-Only/Auditor.
FR-ID-04	The system shall support full CRUD on custom roles per Organization, in addition to the 9 system role templates.
FR-ID-05	The system shall provide a Permission Matrix screen displaying every permission code against every role, with checkboxes editable by users holding role.managePermissions.
FR-ID-06	The system shall enforce every API endpoint against the authenticated user's resolved permission set (union across all assigned roles); a request lacking the required permission shall be rejected with HTTP 403.
FR-ID-07	The system shall enforce that an Evaluator (RE) recording an action's effectiveness verdict is never the same user as that action's Responsible/Owner (RR).
FR-ID-08	The system shall provide a minimal user directory (id, name, email) accessible to any authenticated user for assigning owners/evaluators/team members, independent of the full user.view permission.
FR-ID-09	The system shall issue a signed JSON Web Token (JWT) on successful login, valid for 12 hours, required as a Bearer token on every subsequent API call.

3.5 Governance Settings

Tenant-level configuration of process defaults, KPI thresholds, and alert behavior.

ID	Requirement
FR-GOV-01	The system shall allow an authorized user to configure the default Root Cause Analysis method (5-Why or Ishikawa) for the Organization.
FR-GOV-02	The system shall allow an authorized user to toggle whether a completed REX (E7) entry is required before an NCP Sheet can be closed.
FR-GOV-03	The system shall allow an authorized user to configure the numeric threshold for each of the 10 KPIs.
FR-GOV-04	The system shall allow an authorized user to enable or disable each of the 10 alert types (A-J) independently.
FR-GOV-05	The system shall persist governance settings per Organization and apply tenant-specific defaults automatically on first access.

3.6 License & Plan Management

SaaS/OnPrem deployment tracking and seat/plan administration.

ID	Requirement
FR-LIC-01	The system shall record, per Organization, a plan tier (Starter, Professional, Enterprise) and a deployment model (SaaS or OnPrem).
FR-LIC-02	The system shall track total seats, seats used (computed from active user count), billing cycle, renewal date, and license status.
FR-LIC-03	The system shall allow an authorized user to edit the plan tier, deployment model, seat allocation, and billing cycle.
FR-LIC-04	The system shall recompute seats-used automatically whenever the active user count changes.

3.7 NCP Sheet Lifecycle (E1 through E7)

The central workflow: a structured, staged record of a non-conformity or problem from detection through capitalization, matching KB-003 through KB-009.

ID	Requirement
FR-E1-01	The system shall allow any user holding fiche.create to register a new NCP Sheet capturing title, description, detection date, criticality (High/Medium/Low), priority (P1-P3), frequency (first-time/recurring), and applicable standard.
FR-E1-02	The system shall auto-generate a unique, tenant-scoped Sheet Number in the format NC-{year}-{4-digit sequence} on creation.
FR-E1-03	The system shall set a newly created Sheet's stage to E1 and status to Open.
FR-E1-04	The system shall invoke the Classification Agent on Sheet creation, logging a suggested criticality, suggested priority, and a list of similar past Sheets with a confidence score.
FR-E1-05	The system shall allow the Classification Agent to be re-run on demand from the Sheet's E1 tab.
FR-E1-06	The system shall raise Alert A immediately when a new Sheet is created with criticality High or priority 1.

ID	Requirement
FR-E2-01	The system shall provide a structured 5W2H (What/Where/When/Who/Why/How/How-Much) form for the Problem Understanding stage (E2), plus a free-text Objectives field.
FR-E2-02	The system shall persist exactly one Problem Understanding record per Sheet, editable by users holding fiche.edit.
FR-E3-01	The system shall allow one or more Immediate Actions to be created against a Sheet, each with description, tasks, required means, responsible owner (RR), and planned completion date.
FR-E3-02	The system shall allow the Immediate Action's owner to mark it Done and record the actual completion date.
FR-E3-03	The system shall allow an Evaluator (RE), who must differ from the action's RR, to record an effectiveness verdict (Effective / Not Effective) with comments and efficiency criteria for each Immediate Action.
FR-E3-04	The system shall allow one or more evidence attachments (execution proof / evaluation proof) to be recorded against any action.
FR-E3-05	The system shall invoke the Containment Advisor Agent on request, returning suggested immediate actions drawn from similar closed Sheets.
FR-E3-06	The system shall raise Alert B when an Immediate Action's planned completion date has passed and its status is not Done, and Alert G one day before that date.
FR-E4-01	The system shall allow one or more Root Causes to be recorded against a Sheet, each with a description, a 6M cause category (Man/Machine/Method/Material/Measurement/Milieu), and an RCA method (5-Why or Ishikawa).
FR-E4-02	The system shall allow a Root Cause to be optionally linked to a Standard.
FR-E4-03	The system shall invoke the Root Cause Mining Agent on request, returning suggested root causes and 6M categories drawn from similar closed Sheets.
FR-E4-04	The system shall raise Alert D when 48 hours have elapsed since detection and no Root Cause has yet been recorded for the Sheet.
FR-E5-01	The system shall allow one or more Corrective Actions to be defined per Root Cause, each with description, tasks, required means, responsible owner (RR), and planned completion date.
FR-E5-02	The E5 (Corrective Action Plan) view shall present action definition only; execution status and effectiveness controls shall not be exposed on this tab.
FR-E5-03	The system shall invoke the Action Recommendation Agent on request, returning proven corrective actions for the given root cause(s) drawn from the Capitalization Library.
FR-E6-01	The E6 (Execution & Evaluation) view shall present the same Corrective Actions defined in E5 with execution controls: mark Done, attach evidence, and record an RR/RE-separated effectiveness verdict.
FR-E6-02	The system shall raise Alert C when a Corrective Action is marked Done or its evaluation deadline is 2 days away, and Alert H two days before its planned completion date.
FR-E6-03	The system shall raise Alert I two days before a scheduled evaluation's planned review date.
FR-E7-01	The system shall provide a REX (Retour d'EXpérience) form capturing Lessons Learned, Root Cause Summary, Solution Summary, a Standardization decision (yes/no + details), a Generalization decision

ID	Requirement
	(yes/no + plan), and searchable tags.
FR-E7-02	The system shall invoke the REX Generation Agent on request, auto-drafting the Lessons Learned narrative and standardization/generalization recommendations from the Sheet's full E1-E6 content.
FR-E7-03	The system shall raise Alert E when all Corrective Actions on a Sheet have an Effective verdict, prompting the team to complete E7.
FR-E7-04	The system shall block closing a Sheet (transition to status Closed) unless a REX entry exists, when the Organization's Governance Settings require it (default: required).
FR-E7-05	Upon closure, the system shall set the Sheet's status to Closed, stage to E7, and closure date to the current date, and shall make the Sheet's content available to the Capitalization Library RAG index.
FR-GEN-01	The system shall provide an explicit, sequential stage-advance action (E1->E2->...->E7) gated by fiche.validate, and shall track the Sheet's current stage independently of its status.
FR-GEN-02	The system shall raise Alert J when a Priority-1 Sheet has had no update for 3 or more consecutive days.
FR-GEN-03	The system shall allow assignment of one or more NCP Team Members to a Sheet, distinct from its original detector.

3.8 Action Register

A composable, reusable module underlying both Immediate (E3) and Corrective (E5/E6) actions.

ID	Requirement
FR-ACT-01	The system shall auto-generate a unique, tenant-scoped Action Number in the format A-{year}-{4-digit sequence} on creation.
FR-ACT-02	The system shall track an Action's status through To Do -> In Progress -> Done -> Cancelled.
FR-ACT-03	The system shall provide a personal "My Actions" view listing actions owned by, or pending evaluation by, the authenticated user.
FR-ACT-04	The system shall restrict an action's self-service status update to its assigned owner (or a user holding action.edit).

3.9 Capitalization Library & Retrieval-Augmented Generation (RAG)

The organizational memory of resolved problems, semantically searchable.

ID	Requirement
FR-CAP-01	The system shall automatically index every closed Sheet's title, description, root causes, corrective actions, and REX content into a per-tenant search index upon closure.
FR-CAP-02	The system shall provide a free-text semantic search over the Capitalization Library, ranked by similarity score, restricted to the requesting user's own tenant.
FR-CAP-03	The system shall display, for each closed Sheet in the library, its lessons learned, standardization flag, and generalization flag.

3.10 AI Agents

Eight stage-specific assistants (KB-010) providing suggestions, never autonomous decisions.

ID	Requirement
FR-AI-01	The system shall log every AI agent invocation (agent name, prompt/context, response, confidence score, timestamp, related Sheet) to an immutable AI Agent Log.
FR-AI-02	The system shall never allow an AI agent to autonomously change a Sheet's data; every suggestion shall require explicit human action to apply.
FR-AI-03	The system shall scope every AI agent's retrieval to the requesting user's own tenant (no cross-tenant data in suggestions).
FR-AI-04	The system shall implement the Monitoring & Alert Agent as an always-on process computing Alerts A-J from current data, invocable on demand or on a schedule.

3.11 AI Use Cases Library (with Version Management)

An independent catalog of AI use cases, unrelated to the Capitalization Library, with full version history.

ID	Requirement
FR-AIUC-01	The system shall support full CRUD on AI Use Cases, capturing title, description, sector, business function, AI technique, maturity stage (1-5), status (Idea/Pilot/Production/Retired), owner, expected impact, estimated ROI, and tags.
FR-AIUC-02	The system shall version an AI Use Case's content independently of its metadata; content shall consist of five labeled sections: Inputs, Prompt, Expected Output, Constraints/Guardrails, and Model/Technique Notes.
FR-AIUC-03	The system shall create Version 1 of an AI Use Case automatically upon creation and mark it as the Default and Current version.
FR-AIUC-04	The system shall allow an authorized user to save the current draft as a new version, incrementing the version number and marking it Current; the prior version shall remain in history unmodified.
FR-AIUC-05	The system shall display the full Version History for an AI Use Case, showing version number, author, timestamp, change note, and Default/Current status badges.
FR-AIUC-06	The system shall allow an authorized user to view the full labeled content of any historical version.
FR-AIUC-07	The system shall allow an authorized user to revert to any historical version; reverting shall duplicate that version's content into a brand-new version marked Current, never rewrite or delete history.
FR-AIUC-08	The system shall record who created each version and an optional free-text change note.

3.12 Business Rules

A governed catalog of the process rules that constrain NCP Solver's own behavior, with full CRUD via RBAC.

ID	Requirement
FR-BR-01	The system shall support full CRUD on Business Rules, each with a code, title, rule type (Validation/Workflow/Approval/Naming/Threshold/Escalation), the module it applies to, a Condition (IF)

ID	Requirement
	statement, an Action (THEN) statement, severity (Blocking/Warning/Info), and an owner.
FR-BR-02	The system shall allow a Business Rule to be marked active or inactive without deleting it.
FR-BR-03	The system shall restrict create/edit/delete of Business Rules to users holding the corresponding businessRule permission.

3.13 Controls (COSO Framework)

Internal controls mapped to the five COSO Internal Control - Integrated Framework components, with full CRUD via RBAC.

ID	Requirement
FR-CTL-01	The system shall support full CRUD on Controls, each classified under one of the five COSO components: Control Environment, Risk Assessment, Control Activities, Information & Communication, or Monitoring Activities.
FR-CTL-02	The system shall record, per Control, a control type (Preventive/Detective/Corrective), a testing frequency (Continuous through Annual), an owner, an effectiveness rating (Effective/Partially Effective/Ineffective/Not Tested), last-tested and next-test-due dates, and evidence notes.
FR-CTL-03	The system shall allow filtering the Controls list by COSO component.
FR-CTL-04	The system shall allow a Control to be linked to one or more Risks/Opportunities as a mitigating control.

3.14 Risks & Opportunities

A likelihood x impact risk register covering both risks and opportunities related to operating NCP Solver, with full CRUD via RBAC.

ID	Requirement
FR-RISK-01	The system shall support full CRUD on Risk/Opportunity items, each with a code, title, description, type (Risk/Opportunity), category (Strategic/Operational/Compliance/Financial/Reputational/Technology), likelihood (1-5), impact (1-5), response strategy, mitigation/action plan, owner, and status (Identified/Assessing/Mitigating/Monitoring/Closed).
FR-RISK-02	The system shall automatically compute an Inherent Score (likelihood x impact) and, where residual likelihood/impact are set, a Residual Score.
FR-RISK-03	The system shall allow a Risk/Opportunity to be linked to zero or more mitigating Controls.
FR-RISK-04	The system shall render a 5x5 likelihood x impact heat-map matrix, color-banded from lowest risk (dark green) to highest risk (red), plotting every Risk item in its corresponding cell.
FR-RISK-05	The system shall allow filtering the register by item type (Risk / Opportunity).

3.15 Alerts & Notifications

The ten standard alerts (A-J) from the Knowledge Base, computed by the Monitoring & Alert Agent.

ID	Requirement
FR-ALT-01	The system shall implement all ten standard alerts (A: new high-priority problem, B: immediate action overdue, C: corrective action pending evaluation, D: RCA not started, E: capitalization stage reminder, F: KPI threshold breach, G: immediate action due tomorrow, H: corrective action due in 2 days, I: evaluation due in 2 days, J: priority-1 sheet inactive 3+ days).
FR-ALT-02	The system shall provide an on-demand "Run Monitoring Agent" action, restricted to users holding alert.manage, that (re-)evaluates all alert conditions.
FR-ALT-03	The system shall avoid duplicate alerts of the same type against the same triggering entity within the same day.
FR-ALT-04	The system shall allow a user to mark an individual alert as read.
FR-ALT-05	The system shall display a live unread-alert badge in the navigation sidebar, refreshed periodically.

3.16 Reports & KPIs

The four standard reports and ten core KPIs from the Knowledge Base.

ID	Requirement
FR-REP-01	The system shall provide an Operational NCP Dashboard report listing every open/in-progress Sheet with stage, ageing in days, and immediate/corrective action completion percentages.
FR-REP-02	The system shall provide an Action Plan Monitoring report listing every action with owner, planned date, status, and an overdue flag.
FR-REP-03	The system shall provide a Strategic Problem-Solving Scorecard showing KPIs 2, 5, 7, 8, and 10, a priority distribution chart, and a non-conformity-by-department chart.
FR-REP-04	The system shall provide a Capitalization & Lessons Learned Log report listing every closed Sheet's lessons learned, standardization, and generalization status.
FR-REP-05	The system shall compute all 10 KPIs (see Appendix B) in real time from current data, scoped to the requesting user's tenant.

3.17 Multi-Language & Localization

English, French, and Arabic, with full right-to-left (RTL) layout for Arabic.

ID	Requirement
FR-I18N-01	The system shall provide complete UI translation in English, French, and Arabic for all navigation, labels, form fields, statuses, and messages.
FR-I18N-02	The system shall allow a user to change their display language at any time from the top navigation bar, without reloading lost work.
FR-I18N-03	The system shall persist a user's language preference and apply it automatically on subsequent logins.
FR-I18N-04	The system shall render the entire application right-to-left, including navigation, forms, and tables, when Arabic is selected.
FR-I18N-05	The system shall allow an Organization to configure a tenant-level default language.

3.18 Multi-Tenancy

Complete data isolation between Organizations sharing the same application instance.

ID	Requirement
FR-TEN-01	The system shall scope every data query by the authenticated user's organization_id; no API response shall ever include another Organization's data.
FR-TEN-02	The system shall isolate RBAC configuration, Capitalization Library content, and RAG retrieval per Organization.
FR-TEN-03	The system shall allow multiple independent Organizations, and Organizations grouped under a single Group entity, to coexist on one deployment.

3.19 Audit Trail

An immutable record of every state-changing action, for compliance and traceability.

ID	Requirement
FR-AUD-01	The system shall write an audit log entry for every CREATE, UPDATE, and DELETE operation on a major entity, capturing actor, timestamp, entity type/id, and before/after values.
FR-AUD-02	The system shall make the audit trail viewable, read-only, to users holding audit.view (e.g., the Auditor role).
FR-AUD-03	The system shall ensure audit log writes never block or fail the primary operation they accompany.

4. External Interface Requirements

4.1 User Interfaces

The application presents the following screens, each accessible only when the authenticated user holds the corresponding view permission; navigation itself is filtered by permission (role-based menu visibility).

Screen	Purpose
Login	Authenticate with email/password; language selector; demo-account picker.
Dashboard	Role-aware summary: KPI tiles, recent Sheets, unread alerts, department chart.
NCP Sheets (list)	Filterable/searchable list of all Sheets in scope.
New NCP Sheet	E1 registration form.
NCP Sheet Detail	7 tabs, one per stage E1-E7, each with stage-appropriate AI assistance.
My Actions	Actions owned by, or pending evaluation by, the current user.
Capitalization Library	RAG semantic search over closed Sheets.
Standards	CRUD on the standards knowledge base.
AI Use Cases Library (list)	Card grid of AI use cases with maturity/status badges.
AI Use Case Detail	Metadata editor, current-version labeled-section editor, version history with revert.
Business Rules	CRUD on process business rules.
Controls (COSO)	CRUD on controls, filterable by COSO component.
Risks & Opportunities	Likelihood x impact heat-map matrix + CRUD register.
Reports	4 standard reports in a tabbed view.
Alerts	Alert inbox with manual "Run Monitoring Agent" trigger.
Hierarchy	Group / Organization / Project management.
Organizational Breakdown (OBS)	Site/department/service/team tree management.
Users & Scope	User directory and role assignment.
Permission Matrix	Full role x permission grid editor.
Governance Settings	KPI thresholds, alert toggles, RCA default, REX policy.
License & Plan	Plan tier, deployment model, seats, billing.

4.2 Software Interfaces (REST API)

All interfaces are exposed as a JSON REST API under /api, secured by a JWT bearer token (except /api/auth/login), and enforced per-endpoint by the RBAC permission catalog. The API is grouped by module as follows.

Module	Endpoints
Auth	POST /api/auth/login, GET /api/auth/me, PUT /api/auth/me/language
Hierarchy	GET /api/hierarchy/tree, PUT /api/hierarchy/organization, POST /api/hierarchy/organizations, GET|POST|PUT|DELETE /api/hierarchy/projects
OBS	GET|POST /api/obs, PUT|DELETE /api/obs/:id
Users	GET|POST /api/users, GET|PUT|DELETE /api/users/:id, PUT /api/users/:id/roles, GET /api/users/directory
Roles & Permissions	GET|POST /api/roles, PUT|DELETE /api/roles/:id, GET /api/roles/permissions/catalog, GET /api/roles/permissions/matrix, PUT /api/roles/:id/permissions
License	GET|PUT /api/license
Governance	GET|PUT /api/governance
Standards	GET|POST /api/standards, GET|PUT|DELETE /api/standards/:id
NCP Sheets	GET|POST /api/fiches, GET|PUT|DELETE /api/fiches/:id, POST /api/fiches/:id/transition, POST /api/fiches/:id/close, PUT /api/fiches/:id/understanding, PUT /api/fiches/:id/team, GET|POST|PUT|DELETE /api/fiches/:id/root-causes[/:rcId], PUT /api/fiches/:id/rex, POST /api/fiches/:id/rex/generate-draft
Actions	GET|POST /api/actions, PUT /api/actions/:id, PUT /api/actions/:id/progress, PUT /api/actions/:id/evaluation, GET|POST /api/actions/:id/evidence
AI Agents	POST /api/ai-agents/:ficheId/classification, /containment-advisor, /root-cause-mining, /action-recommendation, GET /api/ai-agents/:ficheId/logs
Capitalization	GET /api/capitalization, GET /api/capitalization/search
Alerts	GET /api/alerts, POST /api/alerts/run, PUT /api/alerts/:id/read
AI Use Cases	GET|POST /api/ai-use-cases, GET|PUT|DELETE /api/ai-use-cases/:id, GET|POST /api/ai-use-cases/:id/versions, GET /api/ai-use-cases/:id/versions/:versionId, POST /api/ai-use-cases/:id/versions/:versionId/revert
Business Rules	GET|POST /api/business-rules, GET|PUT|DELETE /api/business-rules/:id
Controls	GET|POST /api/controls, GET|PUT|DELETE /api/controls/:id
Risks & Opportunities	GET|POST /api/risks, GET|PUT|DELETE /api/risks/:id
Reports	GET /api/reports/kpis, /operational, /action-plan, /scorecard, /capitalization
Dashboard	GET /api/dashboard

4.3 Hardware Interfaces

None. NCP Solver is a standard web application with no direct hardware interface requirement; it runs on commodity server and client hardware.

4.4 Communications Interfaces

- HTTP/HTTPS using JSON request/response bodies for all client-server communication.
- Authentication via a signed JWT Bearer token in the Authorization header, issued at login and valid for 12 hours.
- The web client is served separately from the API and communicates with it exclusively over the REST interface (no server-side page rendering).

5. Non-Functional Requirements

5.1 Performance

ID	Requirement
NFR-PERF-01	Dashboard and list views shall render within 2 seconds under nominal load (per-tenant data volumes as seeded: dozens of Sheets, hundreds of actions).
NFR-PERF-02	AI agent suggestions (classification, containment, root-cause mining, action recommendation) shall return within 5 seconds.
NFR-PERF-03	Capitalization Library search shall return ranked results within 1 second for a tenant's full closed-Sheet corpus.

5.2 Security

ID	Requirement
NFR-SEC-01	All API endpoints except login shall require a valid JWT bearer token.
NFR-SEC-02	Passwords shall be stored only as bcrypt hashes; plaintext passwords shall never be logged or persisted.
NFR-SEC-03	Every authorization decision shall be enforced server-side; client-side permission checks are a UX convenience only, never the security boundary.
NFR-SEC-04	No SQL query shall be constructed by string-concatenating untrusted input; all data access shall use parameterized statements.
NFR-SEC-05	RR/RE segregation of duties (FR-ID-07) shall be enforced server-side, not merely hidden in the UI.

5.3 Availability & Reliability

ID	Requirement
NFR-AVAIL-01	The system shall degrade gracefully if an AI agent call fails: the underlying CRUD operation shall still succeed, with the AI suggestion simply unavailable.
NFR-AVAIL-02	The system shall target 99.9% uptime in a production deployment.

5.4 Scalability

ID	Requirement
NFR-SCALE-01	The architecture shall support horizontal scaling of the web tier independently of the API tier.
NFR-SCALE-02	The system shall support at least 100 concurrent users per tenant without functional degradation.

ID	Requirement
NFR-SCALE-03	The relational schema and RAG index shall each be swappable for a distributed equivalent (e.g., managed PostgreSQL, a vector database) without changing the API contract.

5.5 Usability & Accessibility

ID	Requirement
NFR-UX-01	The UI shall follow the POWERACT Consulting visual identity (color palette, typography, iconography) consistently across all modules.
NFR-UX-02	The UI shall be responsive and usable on desktop and tablet viewport widths.
NFR-UX-03	All interactive controls shall be keyboard-navigable and carry accessible labels (WCAG 2.1 AA target).

5.6 Maintainability & Portability

ID	Requirement
NFR-MAINT-01	The Action Register, Alert & Notification Engine, User Management & RBAC layer, KPI/Reporting engine, and Audit Trail Service shall be implemented as composable modules reusable by other applications (Risk Management, Audit Management, Project Management) via the same API contract.
NFR-MAINT-02	The backend shall run on any standard Node.js 20+ runtime without requiring a dedicated database server, container orchestration, or external API keys.
NFR-MAINT-03	The AI agent implementations shall be isolated behind a stable internal interface so a rule-based implementation can be replaced by an LLM/vector-DB-backed one without changing calling code.

5.7 Auditability & Compliance

ID	Requirement
NFR-COMP-01	Every AI-generated suggestion shall be logged with a confidence score and be attributable to a specific agent and Sheet, per the AIAgentLog entity.
NFR-COMP-02	The Controls module shall support demonstrating COSO Internal Control - Integrated Framework coverage across all five components.
NFR-COMP-03	The system shall retain the full version history of every AI Use Case indefinitely (no hard deletion of prior versions on edit or revert).

5.8 Data Retention & Privacy

ID	Requirement
NFR-DATA-01	The system shall not transmit tenant data to any third-party service by default; AI agent processing shall run entirely within the deployment's own infrastructure.
NFR-DATA-02	The system shall support both SaaS (shared infrastructure, tenant-isolated data) and OnPrem (customer-hosted) deployment models per Organization.

6. Data Model Overview

The relational schema comprises the following core entities (full attribute-level detail is maintained in the Information Model specification and the implementation's schema.sql). All entities carry an organization_id (tenant) foreign key except Group and the global Permission catalog.

Entity	Description
Group	Optional holding entity above Organization; groups sibling Organizations (e.g., a group of companies).
Organization	The tenant boundary. Holds sector, country, and all tenant-isolated data.
Project	Optional sub-entity of an Organization; a Sheet may reference one.
ObsNode	A node in the Organizational Breakdown Structure tree (site/department/service/team).
User	A person; holds RBAC roles; tenant-scoped.
Role	A named set of permissions, tenant-scoped; 9 system templates + custom roles.
Permission	A global catalog entry (module.action) granted to roles.
License	Plan tier, deployment model, seats, billing per Organization.
GovernanceSettings	KPI thresholds, alert toggles, RCA default, REX policy per Organization.
Standard	A normative document (ISO standard, internal procedure).
NCPFiche (NCP Sheet)	The central aggregate: a non-conformity/problem record spanning E1-E7.
NCPTeamAssignment	Join entity: users assigned to a Sheet's team.
ProblemUnderstanding	5W2H structured analysis, one per Sheet.
RootCause	A 6M-categorized cause identified in E4.
Action	Abstract base for Immediate and Corrective actions; owner, dates, status.
ActionEvaluation	Effectiveness verdict for an Action; enforces RR != RE.
ActionEvidence	File attachment evidencing execution or evaluation of an Action.
REXEntry	E7 capitalization content: lessons learned, standardization/generalization.
NotificationAlert	An instance of one of the 10 alert types (A-J).
AIAgentLog	Immutable log of every AI agent invocation with confidence score.
AuditLog	Immutable log of every CREATE/UPDATE/DELETE on a major entity.
AIUseCase	Metadata for an AI use case (stable across versions).
AIUseCaseVersion	A versioned, labeled content snapshot of an AI use case; append-only.
BusinessRule	A governed IF/THEN process rule with severity and owner.

Entity	Description
Control	A COSO-classified internal control with type, frequency, effectiveness.
RiskOpportunity	A risk or opportunity item with likelihood/impact scoring.
RiskControl	Join entity: controls mitigating a given risk/opportunity.

Appendix A — Permission Catalog Summary

The full permission matrix (role x permission) is administered live in the application's Permission Matrix screen and comprises approximately 60 permission codes across 20 modules (dashboard, fiche, action, rootcause, rex, standard, capitalization, user, role, hierarchy, obs, license, governance, aiUseCase, businessRule, control, riskOpportunity, report, alert, audit), each with up to five actions (view, create, edit, delete, and module-specific actions such as validate, close, assignTeam, updateOwn, evaluate, manageRoles, managePermissions, manage, export). Section 2.3 summarizes the resulting access per standard role; the live matrix is the system of record.

Appendix B — KPI Formulas

KPI	Name	Formula	Target
KPI1	Completion Rate - Immediate Actions	(AI Done / AI Total) x 100	100%
KPI2	Effectiveness Rate - Immediate Actions	(AI Effective / AI Done) x 100	>= 85%
KPI3	On-Time Completion Rate	(Actions on time / Total done) x 100	>= 80%
KPI4	Completion Rate - Corrective Actions	(AC Done / AC Total) x 100	100%
KPI5	Effectiveness Rate - Corrective Actions	(AC Effective / AC Done) x 100	>= 80%
KPI6	Completion Rate - Evaluations	(Evals Done / Evals Due) x 100	100%
KPI7	Standardization Rate	(Sheets w/ std. update / Closed) x 100	>= 70%
KPI8	Generalization Rate	(Sheets generalized / Closed) x 100	>= 40%
KPI9	Non-Conformity Count per Department	Count of Sheets per department per period	Trending down
KPI10	Sheet Closure Rate	(Closed / Total Opened) x 100	>= 85%

Appendix C — Alert Catalog (A-J)

Alert	Name	Trigger	Target
A	New High-Priority Problem	New Sheet with criticality High or priority 1	CI Pilot + NCP Team
B	Immediate Action Overdue	Planned date passed, status not Done	RR / CI Pilot
C	Corrective Action Pending Evaluation	Marked Done, or evaluation due in 2 days	Evaluator (RE)
D	RCA Not Started	48h after detection, no root cause recorded	CI Pilot
E	Capitalization Stage Reminder	All corrective actions Effective	NCP Team
F	KPI Threshold Breach	Monthly KPI below configured threshold	Direction + CI Pilot
G	Immediate Action Due Tomorrow	Planned date = tomorrow	Action Owner (RR)
H	Corrective Action Due in 2 Days	Planned date in 2 days	Action Owner (RR)
I	Evaluation Due in 2 Days	Planned review date in 2 days	Evaluator (RE)
J	Priority-1 Sheet Inactive 3 Days	No update for 3+ days on a P1 Sheet	CI Pilot

Appendix D — Glossary

Term	Definition
NCP Sheet	The central record of a non-conformity or problem, spanning stages E1-E7 (referred to as "NCP Fiche" in the original source specification).
E1-E7	The seven stages of the NCP resolution process: Detection, Understanding, Immediate Actions, Root Cause Analysis, Corrective Action Plan, Execution & Evaluation, Capitalization.
RR	Responsable Realisation - the user responsible for executing an action.
RE	Responsable Evaluation - the user responsible for verifying an action's effectiveness; must differ from the RR.
REX	Retour d'EXperience - the structured lessons-learned output of stage E7.
RCA	Root Cause Analysis - the discipline of finding the true underlying cause of a problem (5-Why, Ishikawa).
5W2H / QQOQCCP	Structured problem-description framework: What/Where/When/Who/Why/How/How-Much (French: Qui/Quoi/Ou/Quand/Comment/Combien/Pourquoi).
6M	Ishikawa cause categories: Man, Machine, Method, Material, Measurement, Milieu.
RAG	Retrieval-Augmented Generation - semantic search over the Capitalization Library grounding AI suggestions in the tenant's own historical data.
RBAC	Role-Based Access Control - access rights determined by assigned role(s).
COSO	Committee of Sponsoring Organizations of the Treadway Commission - the internal control framework used to classify Controls.
OBS	Organizational Breakdown Structure - the site/department/service/team tree.
Multi-Tenant	Architecture in which multiple Organizations share one deployment with fully isolated data.
Composable Module	A module (Action Register, Alert Engine, RBAC, Audit Trail) built for reuse by other applications via a shared API contract.